

Algorithms-II (CS304)

Mid Semester Examination, September 2023

Department of CSE, IIT Patna

Total Marks: 60

Duration: 120 min

Important: No doubt will be entertained during the examination

Precise short answers will get better marks

1. Answer the following questions: [4+2+6+3]

- I. Define Interval Scheduling problem. Present an optimal greedy algorithm for that.
- II. Define Independent set and Matroid.
- III. Show that (S, I_k) is a matroid, where S is any finite set and I_k is the set of all subsets of S of size at most k , where $k \leq |S|$.
- IV. What are the two properties of any optimization problem that motivate for a greedy algorithm for that problem? Describe them very briefly (not more than 2 lines for each).

2. Answer the following questions: [3 +4+8]

- I. When should we use Dynamic Programming and when should we go for Divide and Conquer?
- II. What are the differences between Top-down DP (Memoization) and Bottom-up DP (Tabulation)?
- III. A dynamic array is an array with a big improvement: automatic resizing. A full dynamic array creates a new array of twice size and copy all the existing elements in the new array when an append operation is called. Now consider a dynamic array initialized with size 1 and a sequence of m append operations on it. Analyze the amortized cost per operation. Here m is unrestricted.

3. Answer the following questions: [2 +5+4+4]

- I. How can you introduce randomization in your computation?
- II. Describe how to generate Residual graph. Why do we use augmenting path in solving Max-flow problem?
- III. A matching in a Bipartite Graph is a set of the edges chosen in such a way that no two edges share an endpoint. The problem to find a maximum matching in Bipartite graph is called Bipartite matching problem. How can we solve Bipartite matching problem from an algorithm for Max-flow problem?
- IV. Describe contraction algorithm for global min cut problem.

4. Answer the following questions: [3+4+4+4]

- I. Define principal n^{th} root of unity and its properties.
- II. Describe Halving lemma with proof.
- III. What is the upper bound of the maximum degree in a Fibonacci heap? What are the key ideas for proving it (very briefly)?
- IV. What are splaying operation and its different rotation procedures? Describe very briefly.